Alberto Andrés Valdés González.
Degree: Mathematical Engineer.
Work position: ML-Engineer.
Mail: anvaldes@uc.cl/alberto.valdes.gonzalez.96@gmail.com
Location: Santiago, Chile.

Demonstration of Laplace's rule of succession

Setting: The success probability p is unknown, and we take it uniform on $[0,1]$. Given p, the trials $X_1, X_2, \ldots, X_{n+1}$ are independent $Bernoulli(p)$ variables. Let $S_n = X_1 + \cdots + X_n$ be the number of successes in the first n trials.

$$\boxed{P(X_{n+1} = 1 \mid S_n = s) = \frac{s+1}{n+2}} \qquad \Rightarrow \qquad \boxed{P(X_{n+1} = 1 \mid S_n = 0) = \frac{1}{n+2}}$$

Proof:

Given p, the number of successes is binomial:

$$P(S_n = s \mid p) = \binom{n}{s} \cdot p^s \cdot (1-p)^{(n-s)}$$

We will use the Beta function and its relation with the Gamma function:

$$Beta(a,b) = \int_0^1 x^{(a-1)} \cdot (1-x)^{(b-1)} dx = \frac{\Gamma(a) \cdot \Gamma(b)}{\Gamma(a+b)} = \frac{(a-1)! \cdot (b-1)!}{(a+b-1)!}$$

Step 1: probability of the data. Since $p \sim U(0,1)$, its density is $f(p) = 1$ on $[0,1]$:

$$P(S_n = s) = \int_0^1 P(S_n = s \mid p) \cdot f(p)\, dp = \binom{n}{s} \int_0^1 p^s \cdot (1-p)^{(n-s)} dp = \binom{n}{s} \cdot Beta(s+1, n-s+1)$$

$$= \frac{n!}{s! \cdot (n-s)!} \cdot \frac{s! \cdot (n-s)!}{(n+1)!} = \frac{1}{n+1}$$

(Every number of successes $s \in \{0, 1, \ldots, n\}$ is equally likely a priori.)

Step 2: posterior of p. Using Bayes' theorem:

$$f(p \mid S_n = s) = \frac{P(S_n = s \mid p) \cdot f(p)}{P(S_n = s)} = \frac{\binom{n}{s} \cdot p^s \cdot (1-p)^{(n-s)}}{\binom{n}{s} \cdot Beta(s+1, n-s+1)} = \frac{p^s \cdot (1-p)^{(n-s)}}{Beta(s+1, n-s+1)}$$

$$\Rightarrow \; p \mid S_n = s \; \sim \; Beta(s+1, \; n-s+1)$$

Step 3: probability of the next success. Given p, the trial X_{n+1} is independent of S_n, so $P(X_{n+1} = 1 \mid p, S_n = s) = p$. Then

$$P(X_{n+1} = 1 \mid S_n = s) = \int_0^1 P(X_{n+1} = 1 \mid p, S_n = s) \cdot f(p \mid S_n = s)\, dp = \int_0^1 p \cdot \frac{p^s \cdot (1-p)^{(n-s)}}{Beta(s+1, n-s+1)}\, dp$$

$$= \frac{1}{Beta(s+1, n-s+1)} \int_0^1 p^{(s+1)} \cdot (1-p)^{(n-s)} dp = \frac{Beta(s+2, n-s+1)}{Beta(s+1, n-s+1)}$$

$$= \frac{(s+1)! \cdot (n-s)!}{(n+2)!} \cdot \frac{(n+1)!}{s! \cdot (n-s)!} = \frac{(s+1)!}{s!} \cdot \frac{(n+1)!}{(n+2)!} = \frac{s+1}{n+2}$$

$\Rightarrow$

$$\boxed{P(X_{n+1} = 1 \mid S_n = s) = \frac{s+1}{n+2}}$$

Particular case: if no successes were observed in the n trials ($s = 0$):

$$\boxed{P(X_{n+1} = 1 \mid S_n = 0) = \frac{0+1}{n+2} = \frac{1}{n+2}}$$

Interpretation: the estimate behaves as if, before the experiment, we had already seen one success and one failure. Unlike the naive estimate $\frac{s}{n}$, it never returns exactly 0 or 1: after n failures the next success is still possible, with probability $\frac{1}{n+2}$.

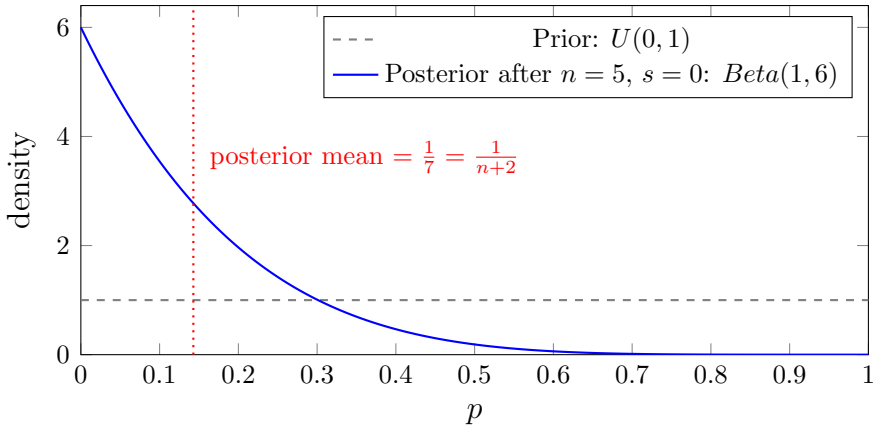


Figure 1: After 5 failures, the posterior concentrates near 0, but its mean is $\frac{1}{7}$, not 0.